codegraph-gitignore-honor-challenge

challenges/scripts/codegraph_gitignore_honor_challenge.sh — BOB-104 §11.4.115 RED/GREEN regression guard

Revision: 1 **Last modified:** 2026-08-19T00:00:00Z **Status:** active **Item:** BOB-104 (CodeGraph 1.5.0 nested-.gitignore honor regression)

Overview

`challenges/scripts/codegraph_gitignore_honor_challenge.sh` is a falsifiable §11.4.115 RED/GREEN polarity guard covering the CodeGraph 1.5.0 regression that walked into `frontend/node_modules` + `extension/node_modules` (both nested-`.gitignore`-excluded), producing 32,260 files / 514,456 nodes vs the 2026-06-06 baseline of 509 files / 8,906 nodes — a ~63x file blow-up. Upstream issue https://github.com/colbymchenry/codegraph/issues/1567; draft PR #1568. Boba's closure obligation is a real, mechanical regression guard on the invariant "a gitignore-honoring indexer must NOT sweep node_modules".

Design (static-scan approach)

The challenge is a **static file-count comparison**, not a live codegraph re-index. Rationale (§11.4.6, weighed against options):

- it does **not** touch `.codegraph/` state;
- it runs in seconds, not minutes;
- it is deterministic and needs no codegraph binary present;

- it is falsifiable — the RED and GREEN thresholds are numeric.

Two populations are measured:

population	command	meaning
correct oracle	`git ls-files -co --exclude-standard | wc -l`	what a gitignore-HONORING indexer would sweep
broken ceiling	`find . -type f -not -path './.git/*' | wc -l`	what a gitignore-BROKEN indexer would sweep

The 2026-06-06 baseline (BOB-104 opening measurement) is **509 files**.

Polarity contract (§11.4.115)

mode	assertion	passes when
`RED_MODE=1`	broken ceiling > BASELINE * 10 (>= 5090)	the risk surface is real (a broken path would definitely blow past any sane baseline)
`RED_MODE=0` (GREEN, default)	correct oracle <= 5000	invariant holds — indexer would not blow up

`GREEN_MAX_FILES = 5000` sits an order of magnitude above the 2026-06-06 baseline, absorbing normal repo growth without becoming decoration. `RED_MIN_BROKEN = 5090` (10 * baseline) proves the guarded risk surface still exists on this checkout (if someone deleted every node_modules the RED assertion would fail honestly — the ceiling would collapse).

Usage

```bash
# GREEN mode (default) — asserts the invariant holds:
bash challenges/scripts/codegraph_gitignore_honor_challenge.sh

# RED mode — asserts the risk surface is real:
RED_MODE=1 bash challenges/scripts/
        codegraph_gitignore_honor_challenge.sh
```

Every run prints the measured baseline / correct-oracle / broken-ceiling / thresholds as evidence (§11.4.6 numbers, not claims).

Exit codes

code	meaning
0	polarity assertion holds for the selected mode.
1	polarity assertion violated (guarded invariant broke).
2	environment error (not a git worktree, tooling missing).

Verification observed 2026-08-19

- GREEN: correct oracle = 2080, threshold 5000 -> exit 0 PASS.
- RED: broken ceiling = 129978, threshold 5090 -> exit 0 PASS.

The correct oracle sits well below GREEN_MAX_FILES; the broken ceiling towers ~63x above it — the exact shape the 1.5.0 regression produced.

Cross-references

- §11.4.6 no-guessing (measured numbers cited)
- §11.4.115 RED-first polarity contract
- §11.4.201 guards assert the real condition (self-falsifiable)
- §11.4.238 automated QA discovers, not confirms
- Upstream: https://github.com/colbymchenry/codegraph/issues/1567, PR #1568